

Set up with Anaconda

For students who already use Anaconda or Miniconda.

Before you start - pilot edition

You need the full course folder, including **environment-anaconda.yml**, and internet for the first setup. Package resolution has been checked on Windows/Python 3.14; fresh-installation and macOS pilots remain to be completed.

This route creates a separate **esbmtk-practicals** environment. You do not need uv or a separate Python installer, and your existing R setup stays unchanged.

1 Open the course folder in a terminal

Extract the downloaded course archive. On Windows, open **Anaconda Prompt** from the Start menu. On macOS/Linux, use a terminal where conda works. Replace the example path with the folder containing **environment-anaconda.yml**.

Windows Anaconda Prompt: /d also changes the drive.

```
cd /d "D:\path\to\ESBMTK-practicals"
```

macOS / Linux:

```
cd "path/to/ESBMTK-practicals"
```

2 Create and activate the environment

Run one line at a time and wait for each command to finish:

```
conda env create -f environment-anaconda.yml
conda activate esbmtk-practicals
python -m pip check
```

The terminal prompt should show **(esbmtk-practicals)**. The last command should report **No broken requirements found**. If this environment already exists, activate and check it instead of creating it again.

3 Register the kernel and launch JupyterLab

```
python -m ipykernel install --sys-prefix --name esbmtk-practicals
python -m jupyterlab
```

Kernel registration is needed only once. It adds the course kernel inside this environment. JupyterLab then opens in your browser; keep the terminal open.

Open your notebooks

Anaconda route / select the kernel, check it, and return next time

4 Select the course kernel

Open **notebooks/student/** in JupyterLab. Select **esbmtk-practicals** in the kernel selector. If the notebook asks for the instructor's ESBMTK314 kernel, choose your course kernel instead. If you previously followed the longer Markdown guide, the same kernel may be labelled **ESBMTK practicals**.

Run a quick check

In JupyterLab, create a new Python notebook with the course kernel. Paste this into a cell and press **Shift+Enter**:

```
import sys
import numpy as np
import esbmtk
import PyCO2SYS

print(sys.executable)
np.testing.assert_allclose(np.linalg.solve(np.eye(2), np.ones(2)), [1, 1])
print("Setup check passed")
```

Expect **Setup check passed**. The printed Python path should contain **esbmtk-practicals**. This checks imports and a real numerical calculation; it does not complete any course exercise.

Restart JupyterLab after shutting it down

Only when JupyterLab is no longer running (for example, after restarting your laptop), open Anaconda Prompt or your Conda-enabled terminal, enter the course folder and run:

```
conda activate esbmtk-practicals
python -m jupyterlab
```

If JupyterLab is already running, open existing notebooks in its file browser or create one via **File > New > Notebook**, then select the course kernel. **No terminal command is needed for each notebook**. Closing a browser tab usually leaves JupyterLab running. Save your work before shutting it down.

If something does not work

conda not found: on Windows, use Anaconda Prompt.

YAML file not found: enter the extracted folder containing environment-anaconda.yml.

Missing module / wrong Python path: activate the environment, launch Jupyter there, and select the course kernel.

Windows Python crash: launch from the activated prompt; send your instructor the error if it persists. Do not reinstall packages at random.

Exercise / NotImplementedError: a marked student placeholder is expected.

Conda help: [Managing environments](#). For other errors, send the complete error message and your operating system to your instructor.